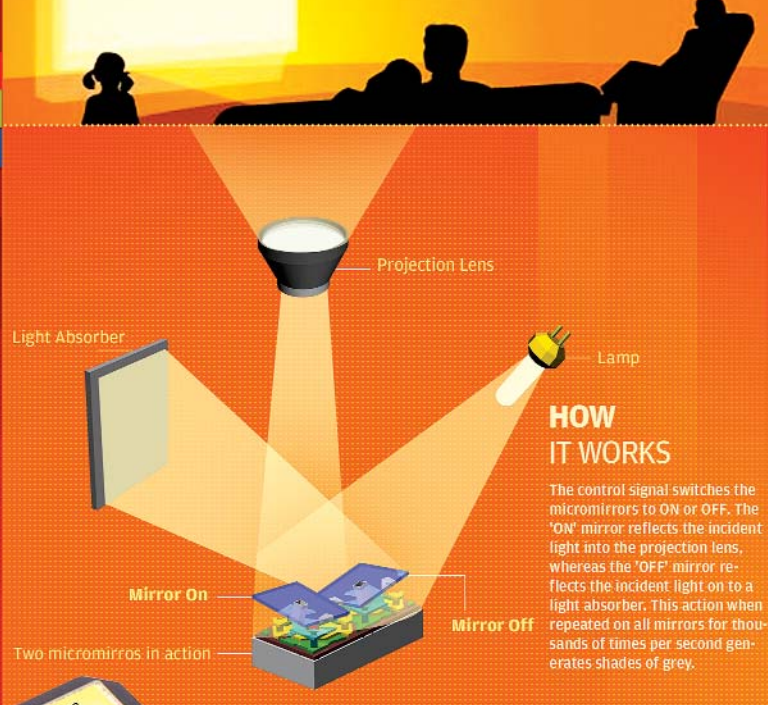
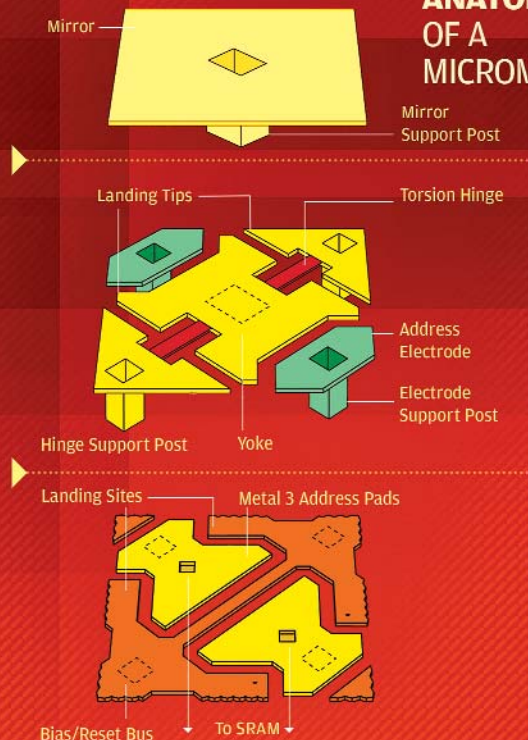


ANATOMY OF A MICROMIRROR



Digital Light Processing

Digital Light Processing, a technology that led to miniaturisation of projectors, has the potential to replace the ageing film-based projection of movies in theatres, and more...

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Digital Light Processing (DLP) is a technique used to project images and is based around a specialised optical semiconductor chip—a Digital Micromirror Device (DMD), developed by Texas Instruments (TI). Today, DLP powers general purpose conference room projectors, and over the years, has spawned a new category of small, ultra-portable mobile projectors.

Although initially developed for projection processes, DLP is now used in telecommunications, scientific instrumentation, volumetric displays, holographic data storages, lithography and medical imaging.



Let's take a closer look at what makes a DLP system tick.

The Chip That Rocks The Cradle

The soul of any DLP system is the Digital Micromirror Device (DMD) chip, which was developed by Dr Larry Hornbeck at Texas Instruments (TI) in 1987. Over the years of development, starting from 1977, to the release of the first commercial product in 1996, DLP technology has been perfected to deliver the goods it promises.

A DMD chip consists of an array of two million microscopic hinged mirrors that can be individually tilted using digital signals. The tilting motion allows them to be used as a switch to modulate light in a particular direction, and are often referred to as light switches or Spatial Light Modulators (SLMs).

Each aluminium micromirror is a 16 micrometer square, polished to reflect light in the desired direction. The combination of a micromirror, the hinge assembly and an address pad forms one pixel of the DMD.

The Stuff It's Made Of

The mirror has a central post on which it is pivoted (as shown above: *Anatomy Of A Micromirror*); the other end of the post is connected to the yoke of the hinge assembly. The hinge assembly consists of a central yoke that is suspended between two support posts via a torsional hinge.

Two address electrodes are placed coinciding with the diagonal ends of the mirror, with a gap of air left between the mirror surface and the address electrode, so as to enable the tilting motion of the mirror.

The hinge assembly, as a whole, is responsible for the tilting motion of the mirror. The